

LEC20 - Binary Search Tree Mutation

Binary Search Tree Definition

- Added ordering conditions to a binary tree
 - Remember a binary tree is a tree that can have at maximum 2 subtrees at each node
 - Values are comparable
 - Values in left subtree are less or equal to root value
 - Values in right subtree are greater or equal to root value
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Binary Search Trees Searching

- Binary search trees make it much easier to find items since they are “sorted”

How to search values in a Binary Search Tree

1. If the tree is empty, return `False`
2. Check root value
 - If target is less than the root, move onto the left subtree
 - Otherwise move onto the right subtree
3. Repeat until you reach a leaf
 - Return whether the root value of the leaf is the desired item
4. Return if an item was found at any of the leaf nodes

```
def __contains__(self, item: Any) -> bool:
    if self.is_empty():
        return False
    elif item == self._root:
        return True
    elif item < self._root:
        return self._left.__contains__(item) # Alternatively <item
in self._left>
    else:
        return self._right.__contains__(item) # Alternatively <item
in self._right>
```

Getting All Items In A Binary Search Tree

```
def items(self) -> list:
    if self.is_empty():
        return []
    else:
        return (
```

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```
self._left.items() +  
[self._root] +  
self._right.items()  
)
```

Representation Invariants of Binary Search Trees

“If `self._root` is not `None`, then `self._left` and `self._right` are `BinarySearchTree` objects”

- If you know that the BST is not empty (`not self.is_empty()`), you never need to check if `self._left` or `self._right` are `None`

Binary Search Tree Deletion

- Takes in the item that we want to delete (if it exists)
- If the item is found then remove the first occurrence of this item from the BST
 - Node gets extracted / removed from the BST
 - 3 cases
 1. Node is a leaf (no children)
 - Set `self._root`, `self._left`, `self._right` to `None`
 2. Node has one child
 - Promote the one subtree
 3. Node has two subtrees (more complex)
 - BST properties must still hold after deletion
 - i. Every node in the left subtree must be smaller than the root, and every node in the right subtree must be greater than the root
 - ii. Solution is to promote the max value from the left subtree, OR the min value from the right subtree
- If the item is not found, then do nothing